

Preparing for the open-book coding challenge

Applied Data Science 2, semester 2

Dmytro Shytikov

2026-05

Lecture outline

- GENERAL EXPLANATIONS
- GETTING READY FOR THE EXAM
- GENERAL RECOMMENDATIONS
- FINAL REMARKS

Learning outcomes

At the end of this lecture you should be able to:

- Understand how to prepare for the final open-book coding challenge;
- Understand what we expect you to do;
- Understand how the marking is done;
- Improve study skills;
- Practice open-book assessments.

GENERAL EXPLANATIONS

What is the coding challenge and why are we doing this?

The coding challenge includes:

- 3 problems (not just coding!); 25% each
- document everything in R Markdown 25%
- 3 hours in total
- open-book format with no restrictions.

Why are we doing this?

Assessments are most effective if they are similar to what you would be doing “in real life”

Packages that are available on the computers in the computer class

- all basic packages
- `tidyverse`
- `rmarkdown`
- `knitr`
- `tinytex`
- `pwr`

All tasks can be solved using just these basic libraries.

Also, you should be able to install packages from the RStudio CRAN server.

GETTING READY FOR THE EXAM

We expect you are familiar with

Topics you studied during this semester

- Linear regression models – handle linear regression models, run parametric/non-parametric correlation
- Analysis of variance – perform parametric and non-parametric ANOVA, simple factorial ANOVA, post-hoc tests
- Bootstrapping – run re-sampling using R
- Analysis of qualitative data – perform a χ^2 test
- Computer modeling – not included in the exam
- Time series analysis – not included in the exam
- Cluster analysis – be able to do the basic approaches in R
- Basics of conditional probability and Bayes' stats
- Analyzing categorical data.

Topics from semester 1:

- General skills in R and R Markdown – you can work in RStudio, **you can load files into R**, you can knit an R Markdown file at least *somehow*
- You can visualize your data
- You can formulate statistical hypotheses – no differences between the experimental groups VS there are some differences between the groups
- You have some ideas about how to plan an experiment, what possible pitfalls are, and how to perform power analysis

Look for the clues!

- **Import the data and clean it to remove any inconsistencies (if they are available).** – there may be some problems with data cleaning.
- **Plot it in a way that addresses the question we are interested in.** – build a plot
- **Choose the appropriate statistical test and explain your choice, formulate the correct hypotheses, and check assumptions for this test.** – do exactly as you are told and explain these steps explicitly
- **... and name at least N possible ways in which the study could be improved or followed up on.** – it is a request to show your creativity and general understanding of biology. Do not ask for a bigger sample size unless you can explain why you need it (low power of the current study? How much low? Show the power) and how much you would like to increase it.
- **... There is no need to *do something*.** or **... Do this analysis in a *certain way*.** – we want you to do something in a certain way. Please, follow this request.

sample size can be used, but be cautious; and show power of study

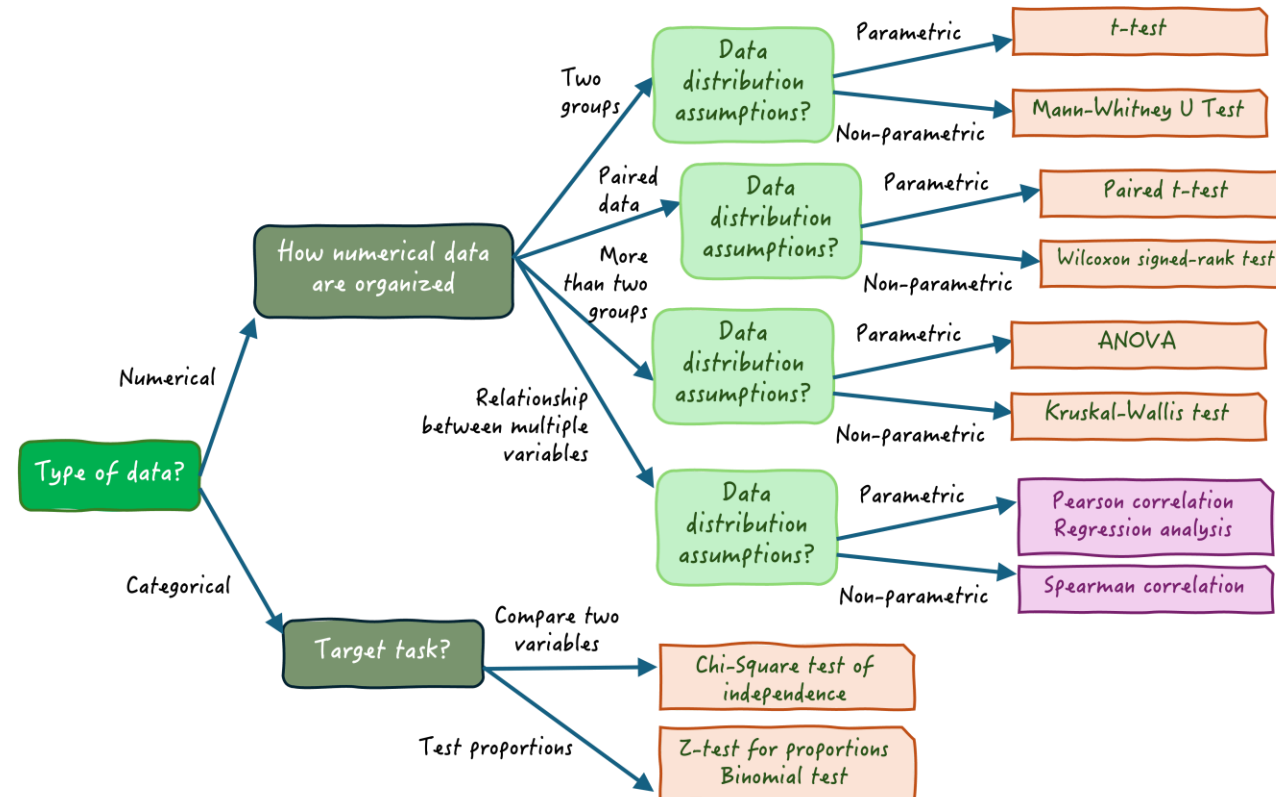
GENERAL RECOMMENDATIONS

How to prepare for an open-book assignment?

You can look up anything, but you cannot look up everything!

- Have your code and lecture materials/notes where you can find them;
- Label code that did and didn't work;
- Write good data-wrangling maneuvers. For instance, **you must be able to load your file into R**;
- Make a hand-written “cheat sheet”, e.g. of frequently-used commands, statistical tests, and their assumptions, and plots;
- Bookmark what you can find in each of the allowed resources (remember to cite when using in assessment) ;
- Make sure you are familiar with the assessment instructions (check Blackboard Learn)

What test to choose?



Choosing the right statistical test for the appropriate scenario

How to describe continuous data

Variables	Plot
Independent: categorical	Boxplot
Data: unpaired	Whisker plot (+SD/SE/CI/IQR)
	Strip chart
Independent: categorical	All above with the primary points that are linked
Data: paired	Histogram of the <i>difference</i>
Dependent: continuous	2D scatter plot or similar
Independent: continuous	A trend line would be helpful

How to describe categorical data

Variables	Plot
Independent: categorical	Bar chart
	Table (with percentages)
	Pie chart

Why do we want you to write statistical hypotheses?

Solving questions in statistics depends on how you formulated your question.

We want you to demonstrate that you understand *how* the respective test works, but not do it mechanically just because you heard it is used under these circumstances.

Some approaches (bootstrapping) actually depend on the way how you formulate your task → these methods *require* you to understand what you are doing.

In some cases, better results can be obtained if you use the one-sided hypothesis vs the two-sided one *correctly*.

Just state the hypotheses correctly if you are asked to do so. (If not, still better do so.)



How to discuss results?

- Formal summary of your results with some statistics, p-value (but **not** only the p-value), possibly, effect size estimate, and your conclusion. Consider the following links:
 - [How to report t-test results](#);
 - [How to report ANOVA results](#).
- What do these data mean? Why do you think it means something good/bad?
- Are there any shortcomings in the study (apart from the sample size)? Was something not done? Is the mechanism of the effect clear? Are all the clinical covariates included in the explanation?
- Is the power of the study good enough? No? Why do you think so? Can you say how many replicates to add? *Power analysis would make your words more convincing!*



www.uk.assignmentgeek.com

Power analysis

Important steps

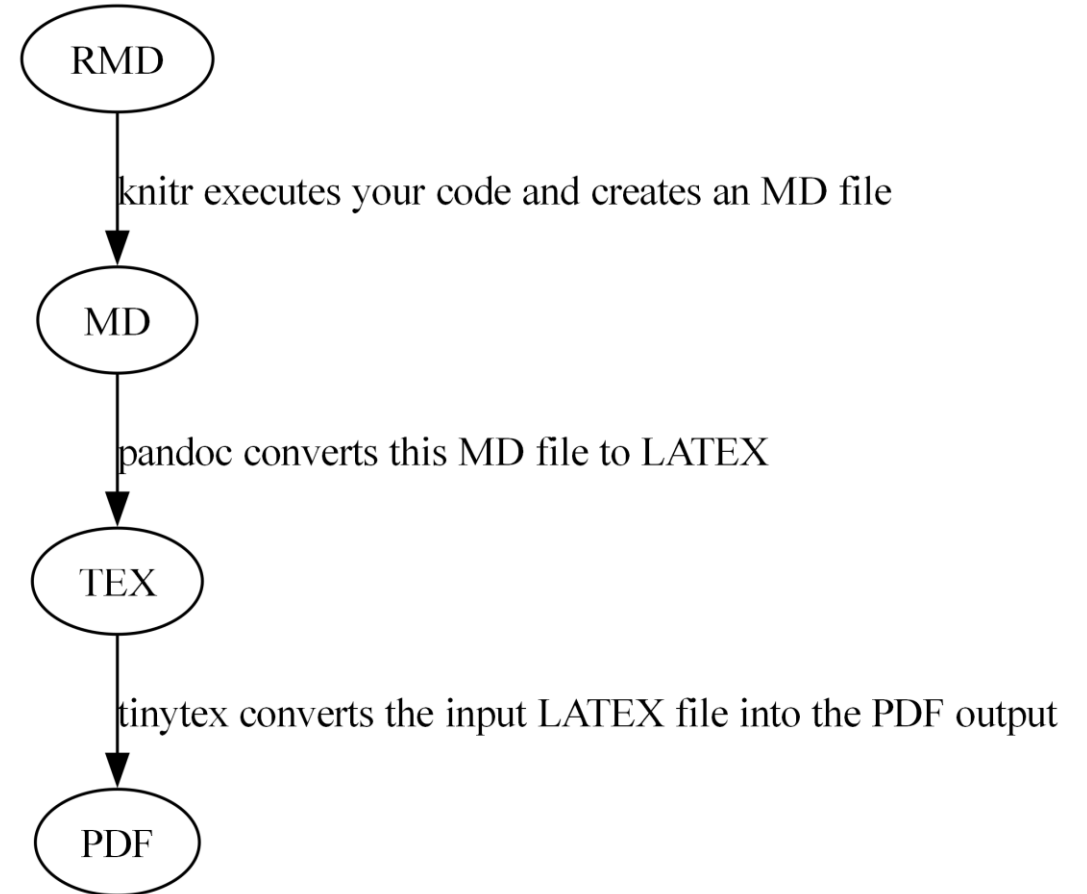
- Find out the power of the study;
You must know the current sample size, effect size, significance level, and test type.
- If the power is high enough (≥ 0.8), your study likely has a sufficient sample size $\rightarrow$ you likely do not need more.
- Otherwise (H_0 is preserved and the power is ≤ 0.8), set the desired test parameters and get the required sample size.

Are you sure that you need more test subjects? What will happen if you get a lower p-value?

Learn to knit!

- Prepare a list of useful commands for `knitr` code chunks, YAML headers, and LATEX.
- Be aware of the limitations of R Markdown and LaTeX.
- Make use of the Markdown template you are provided with.
- Do not copy text from external sources right away.
- As a rule of thumb, it is usually enough to not try anything strange with Markdown and just keep it as simple as possible.

However...



Got troubles producing a PDF file?

- Check the error message carefully. Mistakes may happen because of
 - `knitr` (errors in your code)
 - LATEX (errors in the document formatting, inappropriate symbols, etc)
- Make sure that you can produce *any* R Markdown output (that `knitr` works at all).
PDF did not work? Try HTML or DOCX. Still better than nothing.
- Check whether the `.MD` and `.TEX` files are generated at all (set `keep_md: true` or `keep_tex: true` in the RMD file header).
Cut your documents into chunks and try to knit them in order to locate the error.
- Do not try to play with LATEX and `tlmgr`. It is a bad idea during the exam. Simplify your document instead.
- Notify me. I will try to help if the package does not work. At least, I can note that there was a technical problem. But it does not mean that I will automatically agree or even my opinion will be accepted by the exceptional circumstances committee.
Quick note: Sometimes, the `xfun` package does not work well. In this case, uninstall it from your computer and install again. It takes a little of time and it is likely what I am going to do anyway.
- *In the worst case*, submit whatever you have (RMD? R? Any other output?).

File formatting

Bad formatting

```
#header of section 1
## Subheader 1
Some text
```{r}
some code
```

more text and comments
## Subheader 2
etc ...
```

The output will be:

#header of section 1 ## Subheader 1 Some text
and so on...

Solution:

Add lines between each element of your document

Normal formatting

```
# Header of section 1

## Subheader 1

Some text

```{r}
some code
```

more text and comments

## Subheader 2

etc ...
```

FINAL REMARKS

Conclusions

Now, you should understand:

- what we *likely* look at in your submission;
- what *likely* to expect during the exam;
- what *likely* to use in some situations;
- how to prepare for the exam.

Further suggestions

```
data() # check in-built R datasets

data("Indometh") # paired data ANOVA
data("ToothGrowth") # 2-way factorial ANOVA
data("diamonds") # many-many things...
data("cars") # regression and correlation
data("iris") # many-many approaches...
library("datarium") # Includes many useful datasets for practice
```

Or you can check for some data online.

THANK YOU FOR
ATTENTION!